

New features for GRNsight: a web application for visualizing models of small- to medium-scale gene regulatory networks

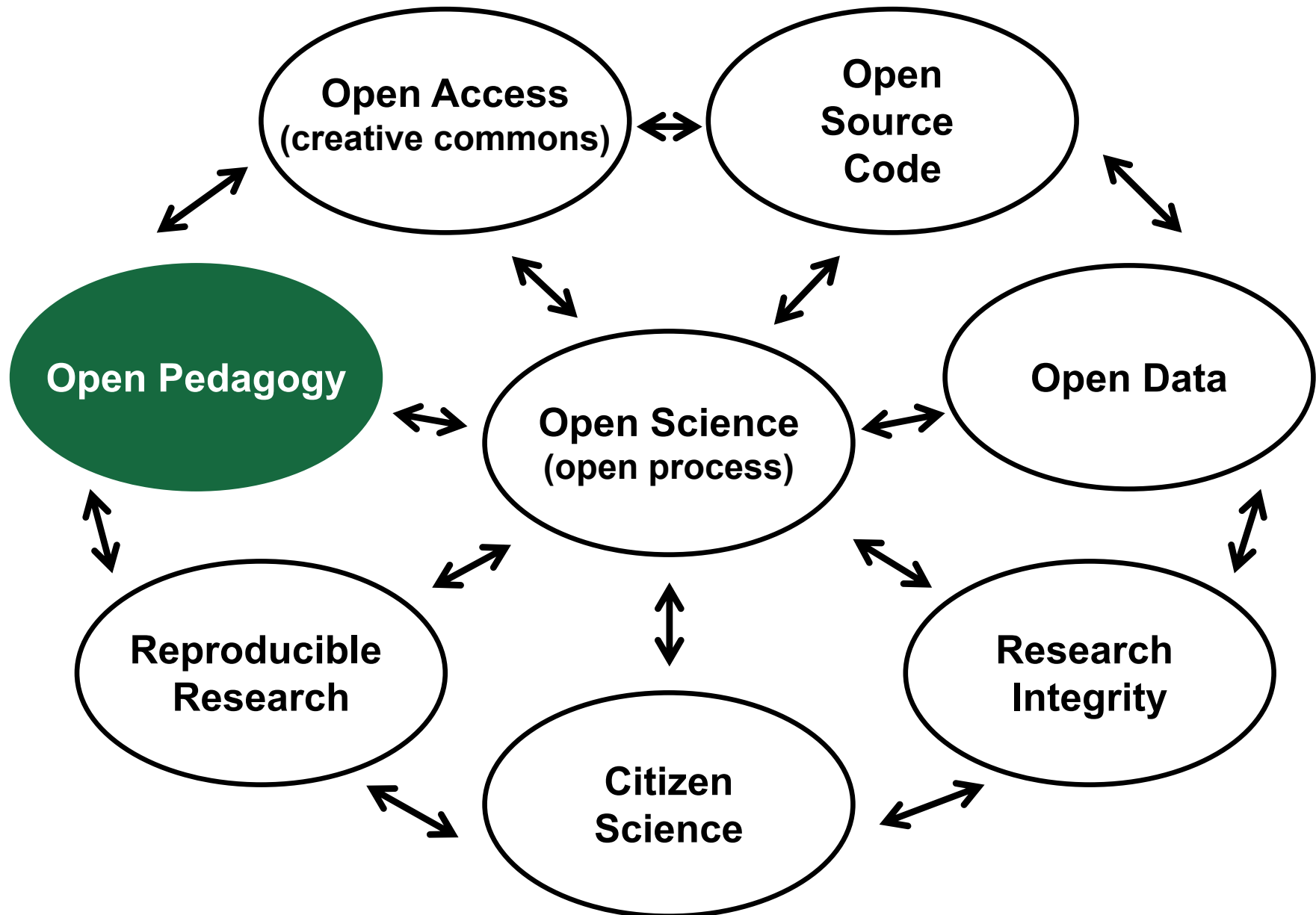
Kam D. Dahlquist, Ph.D.
Department of Biology
Loyola Marymount University

Bioinformatics Community Conference
Bioinformatics Open Source Conference
July 19-22, 2020



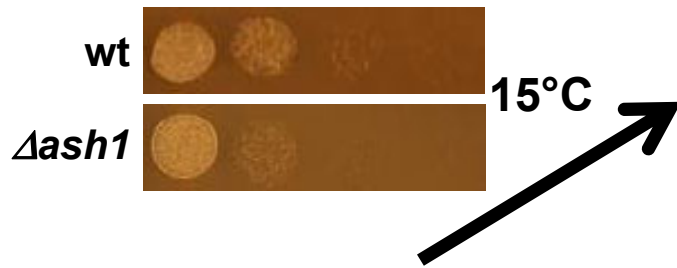
Loyola Marymount University
Frank R. Seaver
College of Science
and Engineering

Open Science Ecosystem

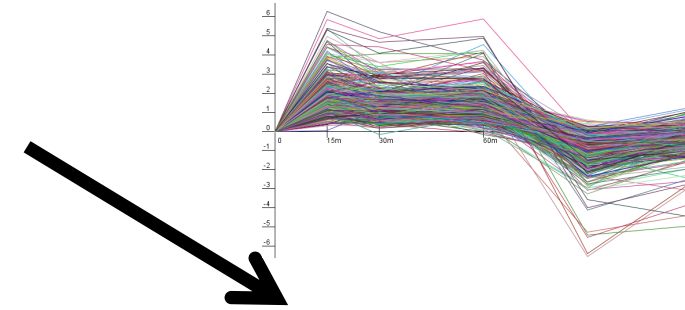
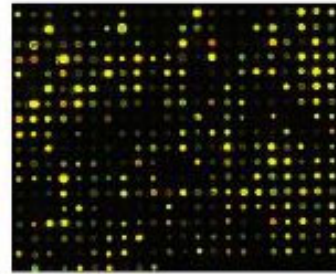


With thanks to John Jungck

GRNsight Is Developed and Used by Undergraduates in Courses and an Interdisciplinary Research Team



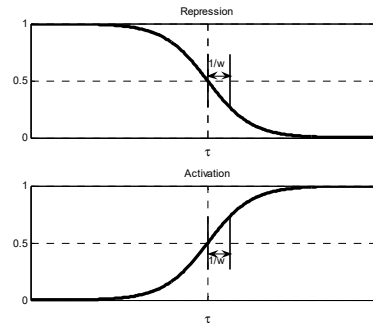
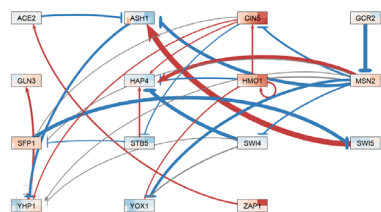
Cold shock microarray data from wt and TF deletion strains



Interpretation,
new questions,
new experiments

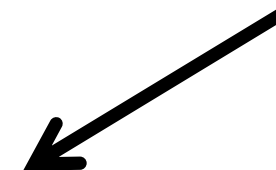
Normalization,
statistical analysis,
clustering

Visualization of
modeling results
using GRNsight



Dynamical systems
modeling using
GRNmap

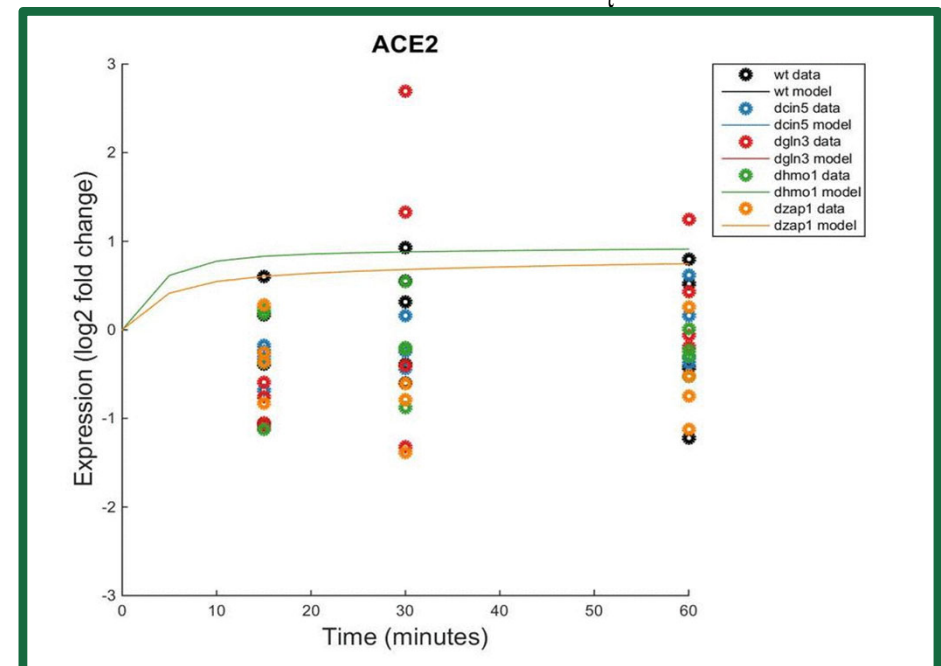
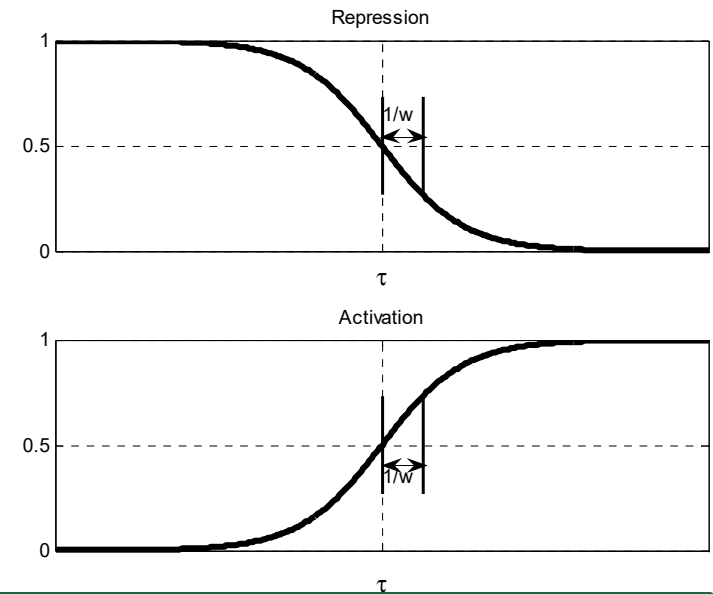
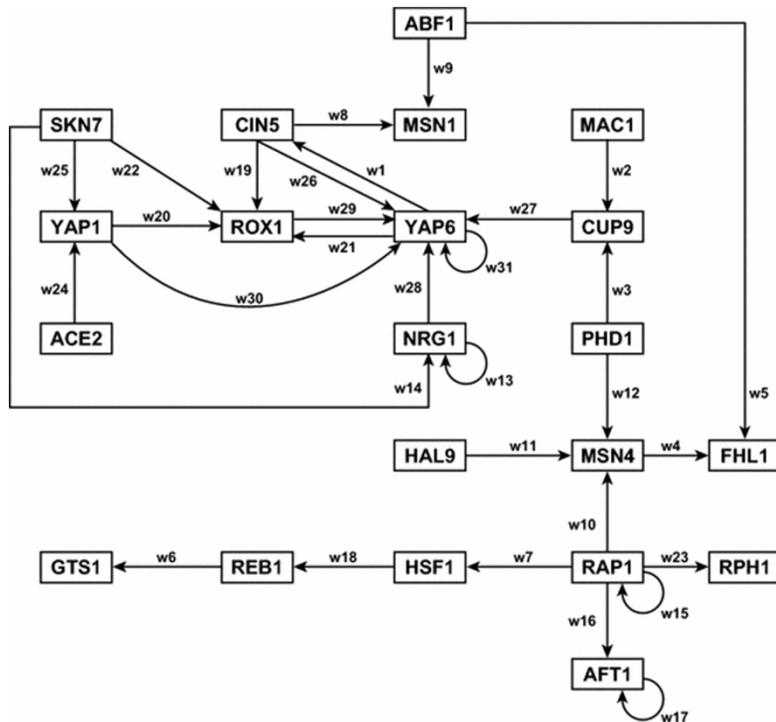
Derivation of gene
regulatory networks
from YEASTRACT



GRNmap: Gene Regulatory Network Modeling and Parameter Estimation

$$\frac{dx_i(t)}{dt} = \frac{P_i}{1 + \exp\left(-\left(\sum_j (w_{ij} x_j(t)) - b_i\right)\right)} - d_i x_i(t)$$

Weight parameter, w , gives the direction (activation or repression) and magnitude of regulatory relationship.



GRNsight Rapidly Generates GRN Graphs Using Our Customizations to the Open Source D3 Library



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Last modified:
6/11/2018

GRNsight v3.1.3

Welcome to GRNsight. Please select a file to upload.

FileViewLayoutNodeEdgeHelpDemo15-genes_28-edges_db5_Dahlquist-data_input.xlsx15 nodes28 edges

Grid Layout

Node Coloring

Enable Node Coloring

Top Dataset

wt_log2_expression

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value

(-100 - 100): 3Set

Link Distance (1 - 1000): 500

Charge (-2000 - 0): -50

☐ Lock Force Graph Parameters

Reset Force Graph Parameters

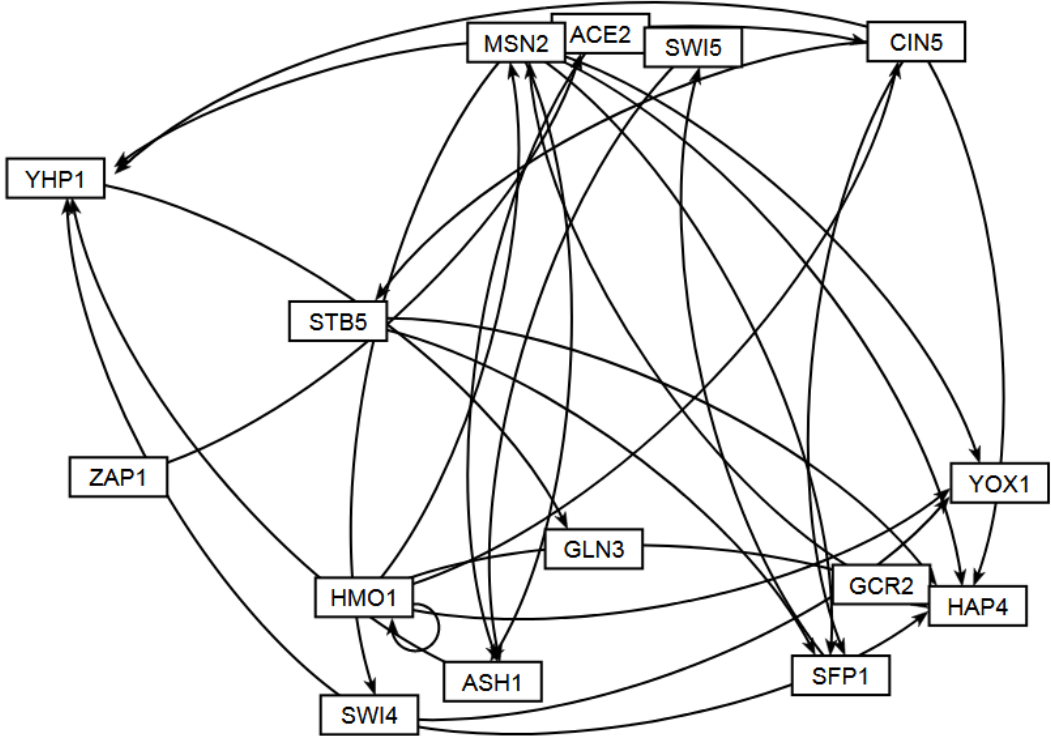
Undo Reset

☒ Restrict graph to viewport

Viewport Size

Small (1104 X 648 pixels)

Zoom (1 - 200%): 100%



Besides the Force Graph Layout, Users Can Rearrange the Nodes by Hand



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File View Layout Node Edge Help Demo 15-genes_28-edges_db5_Dahlquist-data_input.xlsx 15 nodes 28 edges

Force Graph

Node Coloring

Enable Node Coloring

Top Dataset

wt_log2_expression

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value

(-100 - 100): 3 Set

Link Distance (1 - 1000): 500

Charge (-2000 - 0): -50

☒ Lock Force Graph Parameters

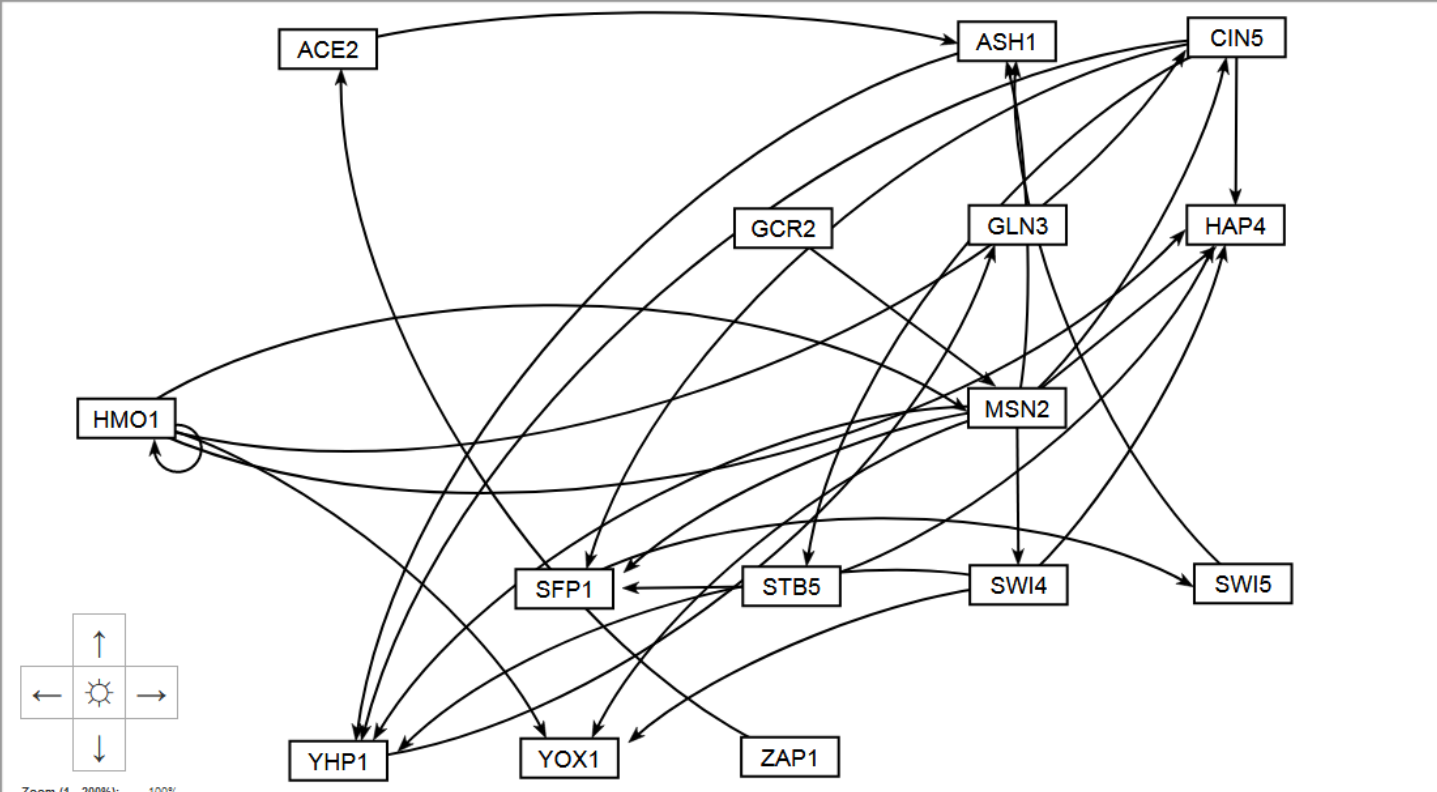
Reset Force Graph Parameters

Undo Reset

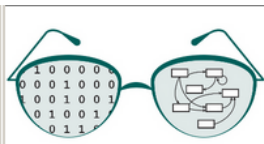
☒ Restrict graph to viewport

Viewport Size

Small (1104 X 648 pixels)



Or Choose the Convenient Block Layout Option



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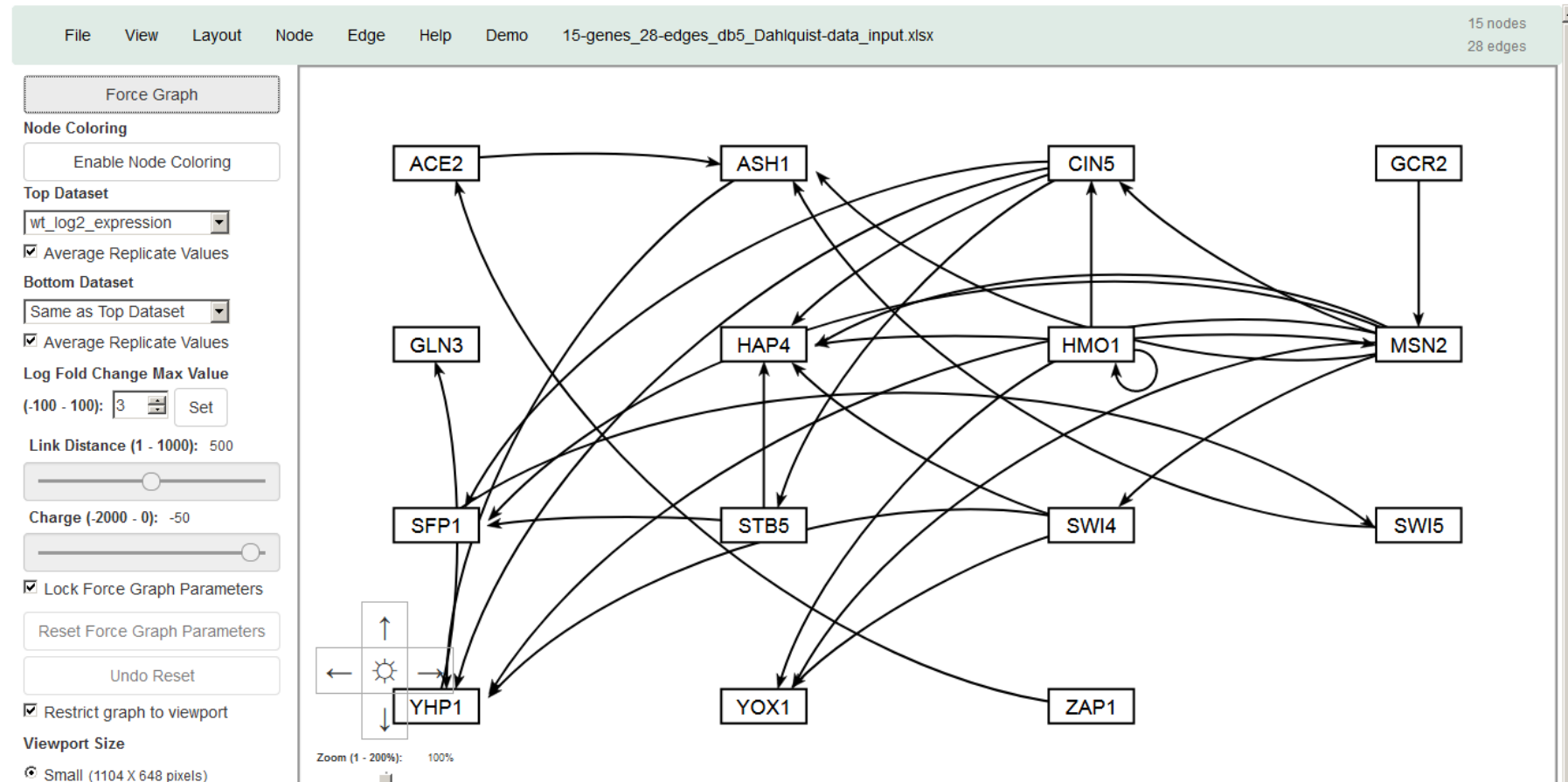
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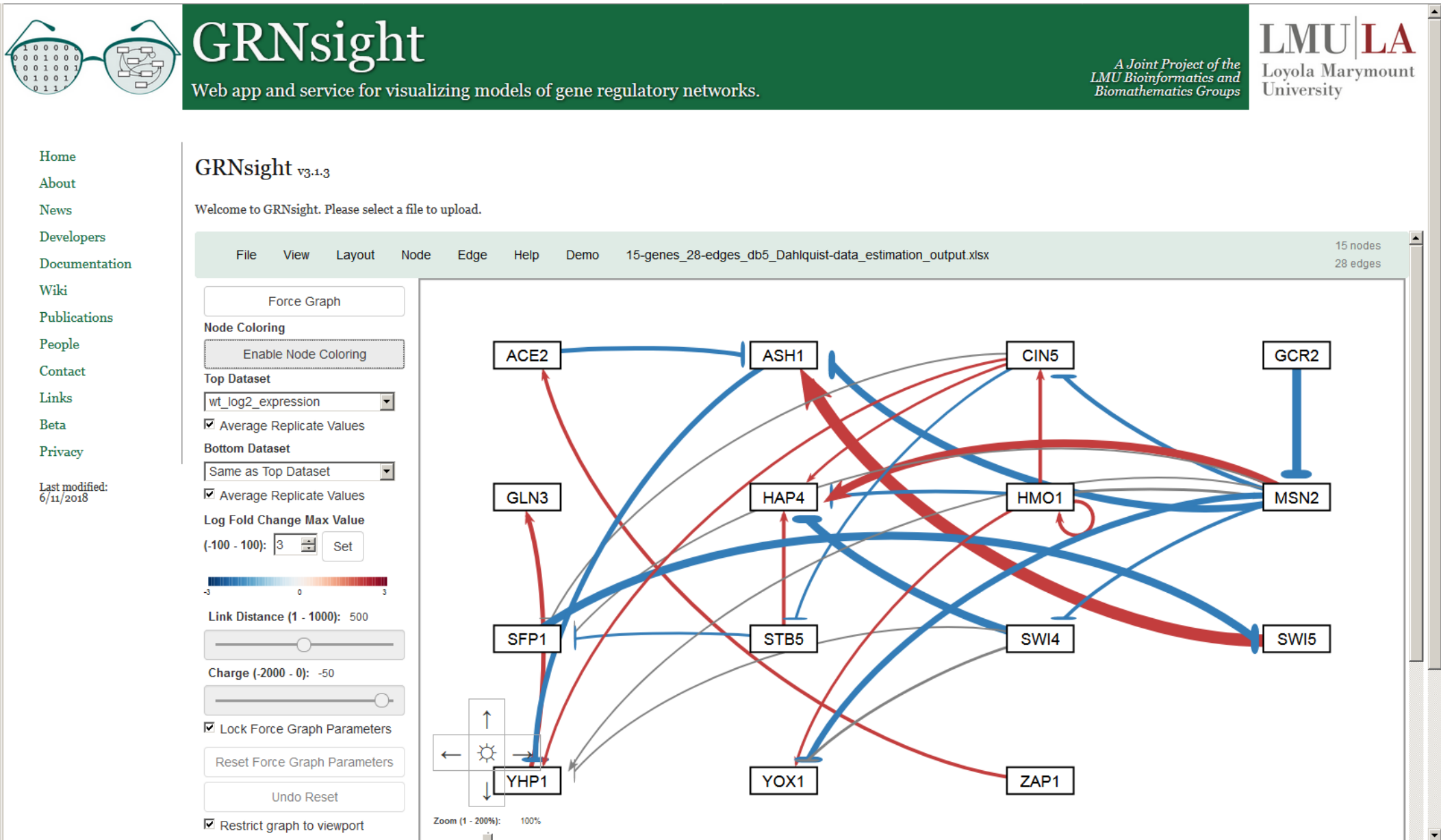
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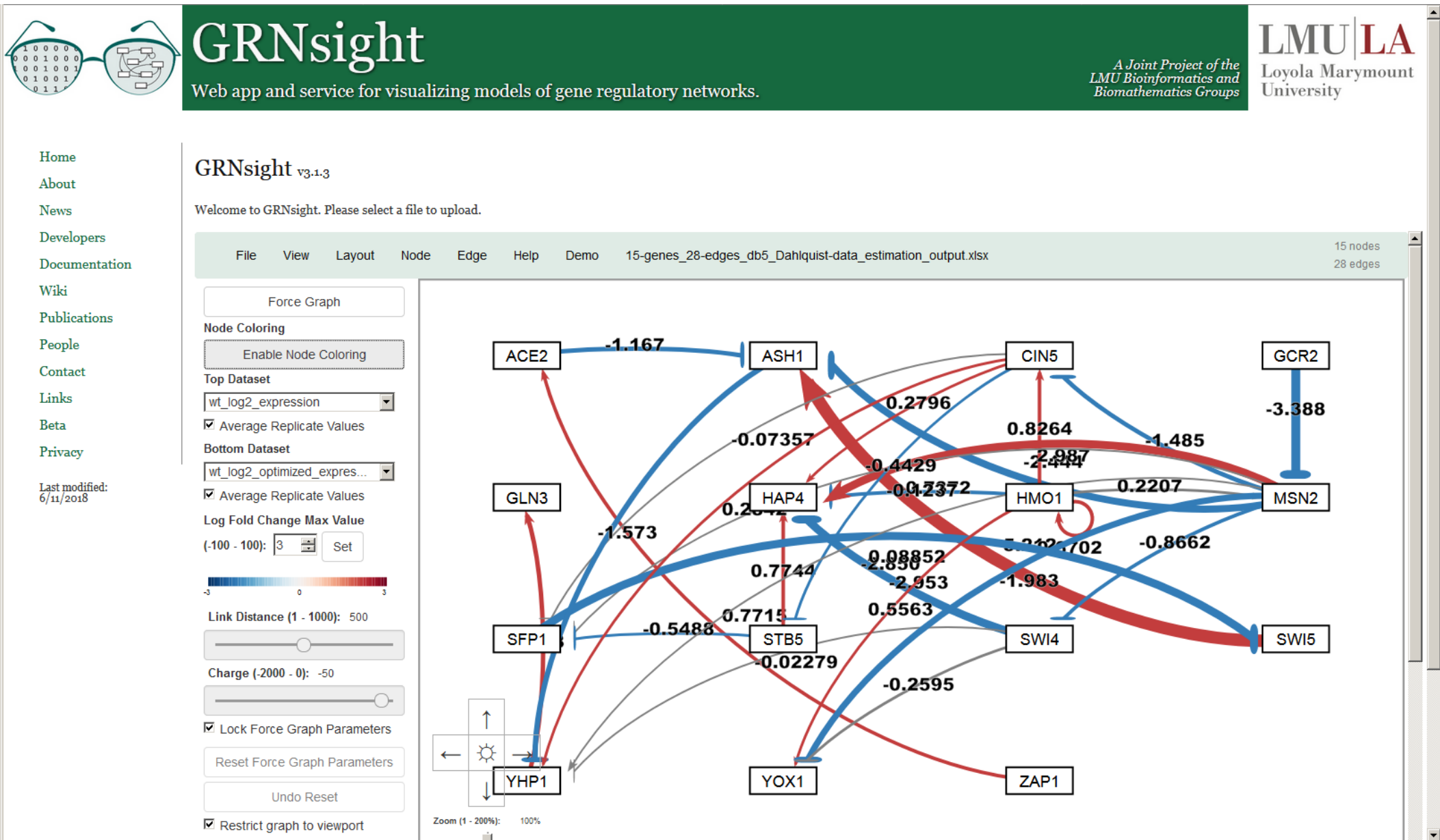
New in Version 3

When Model Results are Loaded, Edge Weight Values Are Indicated by Color, Thickness, and End Marker



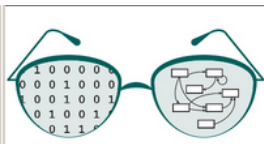
New color scheme in Version 3

Numerical Weight Values Can Be Displayed on the Graph



New in Version 2

Edge Thicknesses are Normalized to the Largest Magnitude Weight Value



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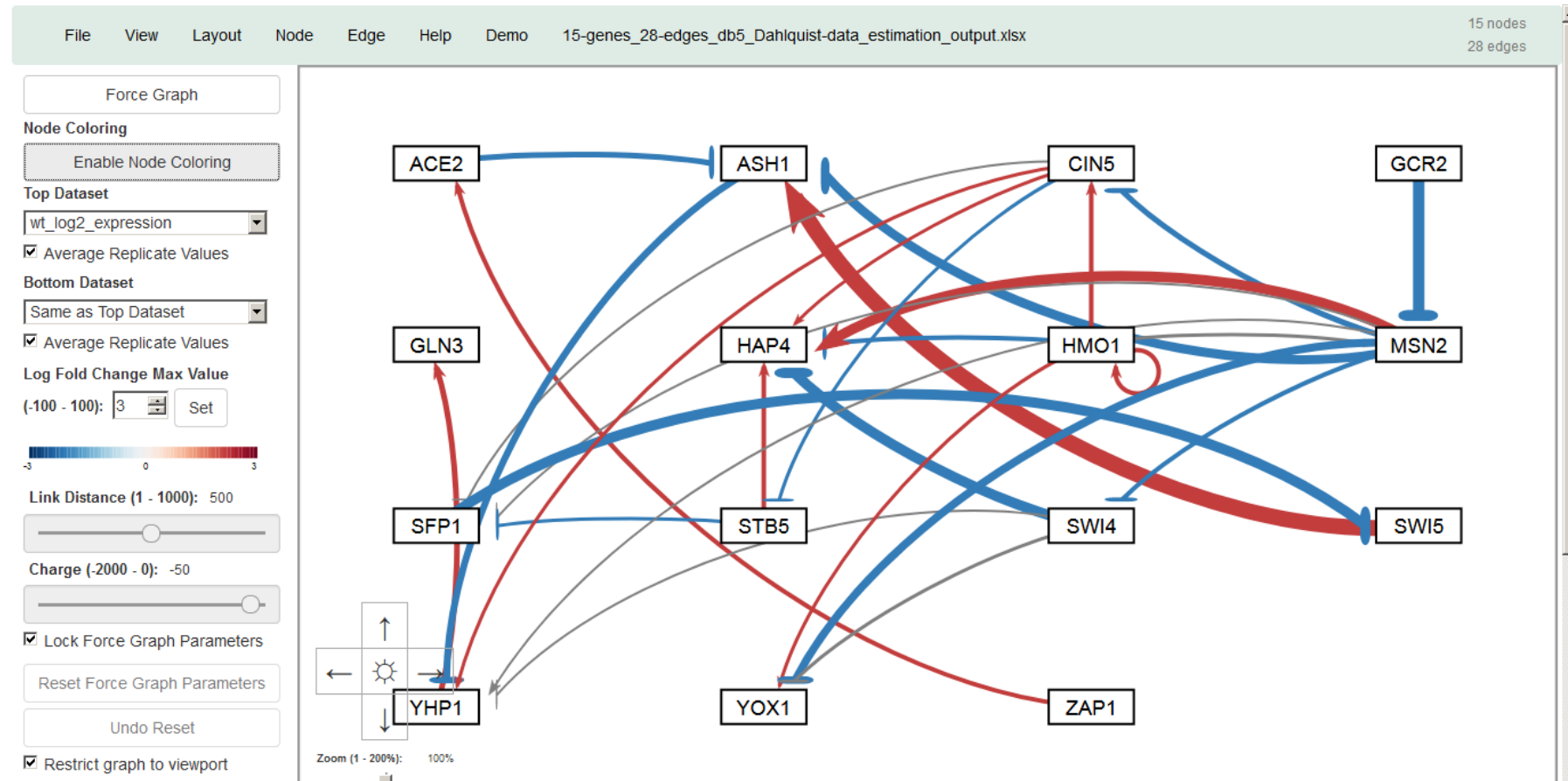
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GRNsight v3.1.3

Normalization Factor = 5.21

Welcome to GRNsight. Please select a file to upload.



Version 1

And Can Be Modified to Allow Fair Visual Comparison between Different Graphs



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GRNsight v3.1.3

Welcome to GRNsight. Please select a file to upload.

Normalization Factor = 10

File View Layout Node Edge Help Demo 15-genes_28-edges_db5_Dahlquist-data_estimation_output.xlsx 15 nodes 28 edges

Force Graph

Node Coloring

Enable Node Coloring

Top Dataset

wt_log2_expression

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value

(-100 - 100): 3 Set

Link Distance (1 - 1000): 500

Charge (-2000 - 0): -50

☒ Lock Force Graph Parameters

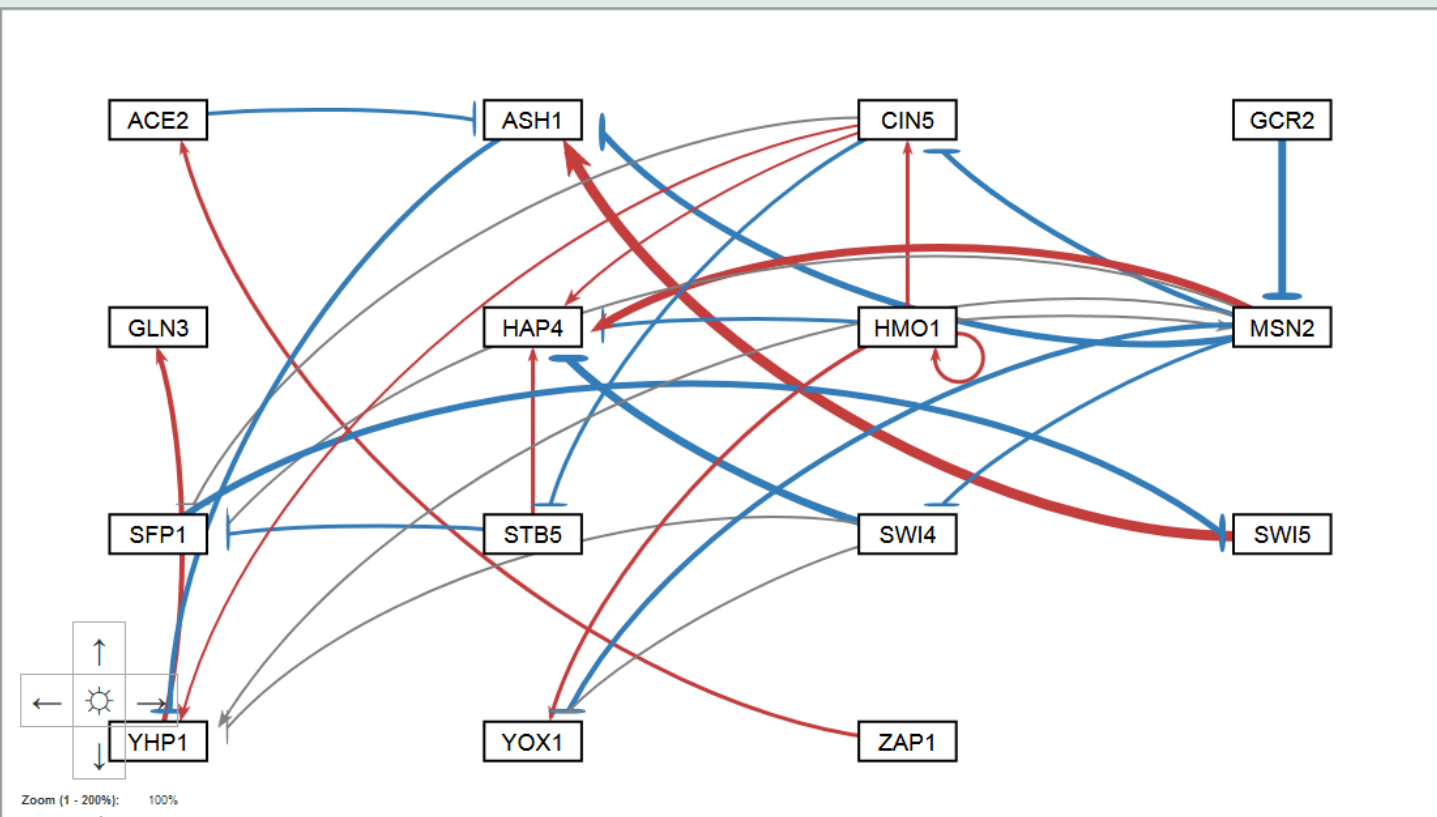
Reset Force Graph Parameters

Undo Reset

☒ Restrict graph to viewport

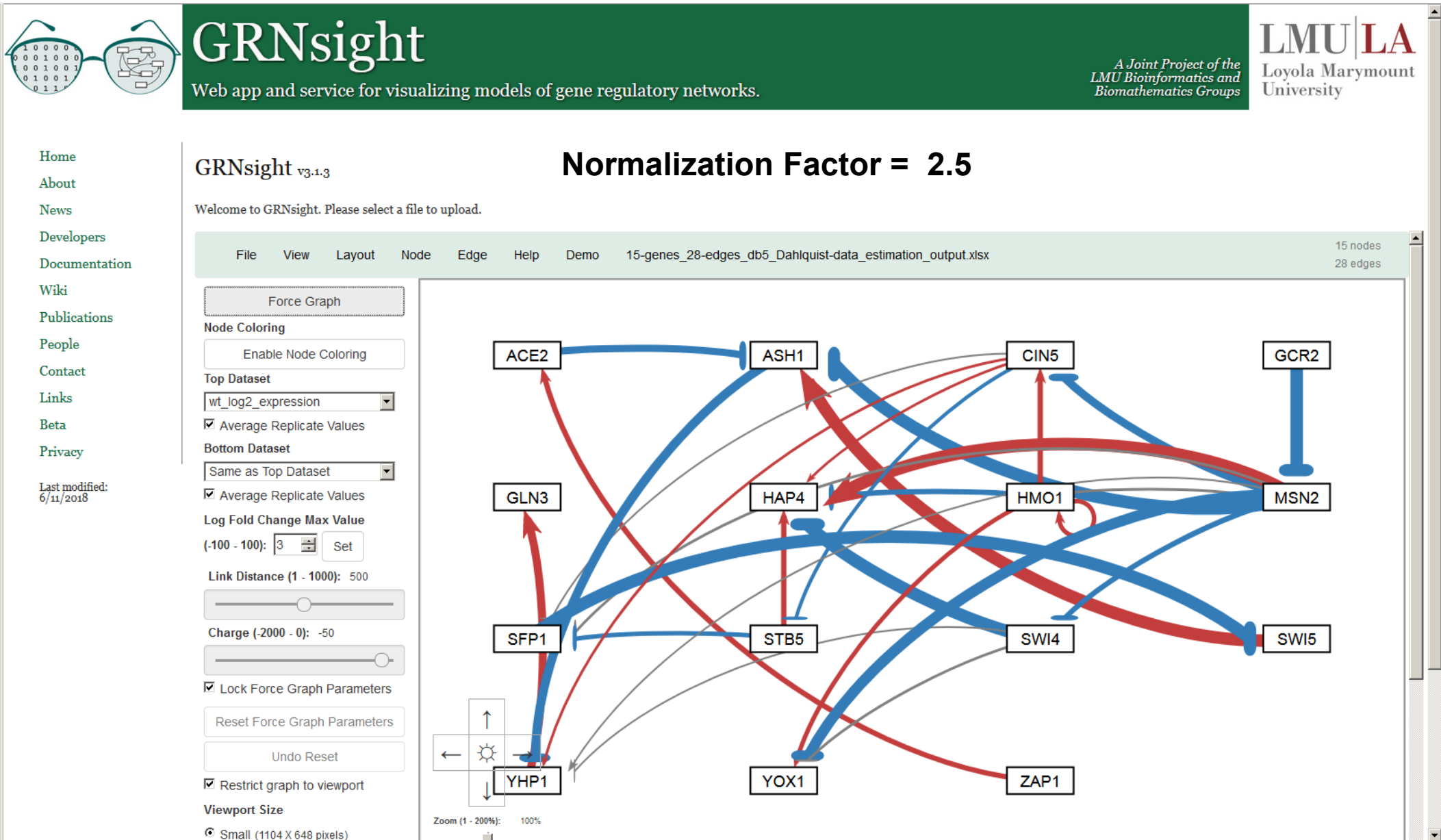
Viewport Size

Small (1104 X 648 pixels)



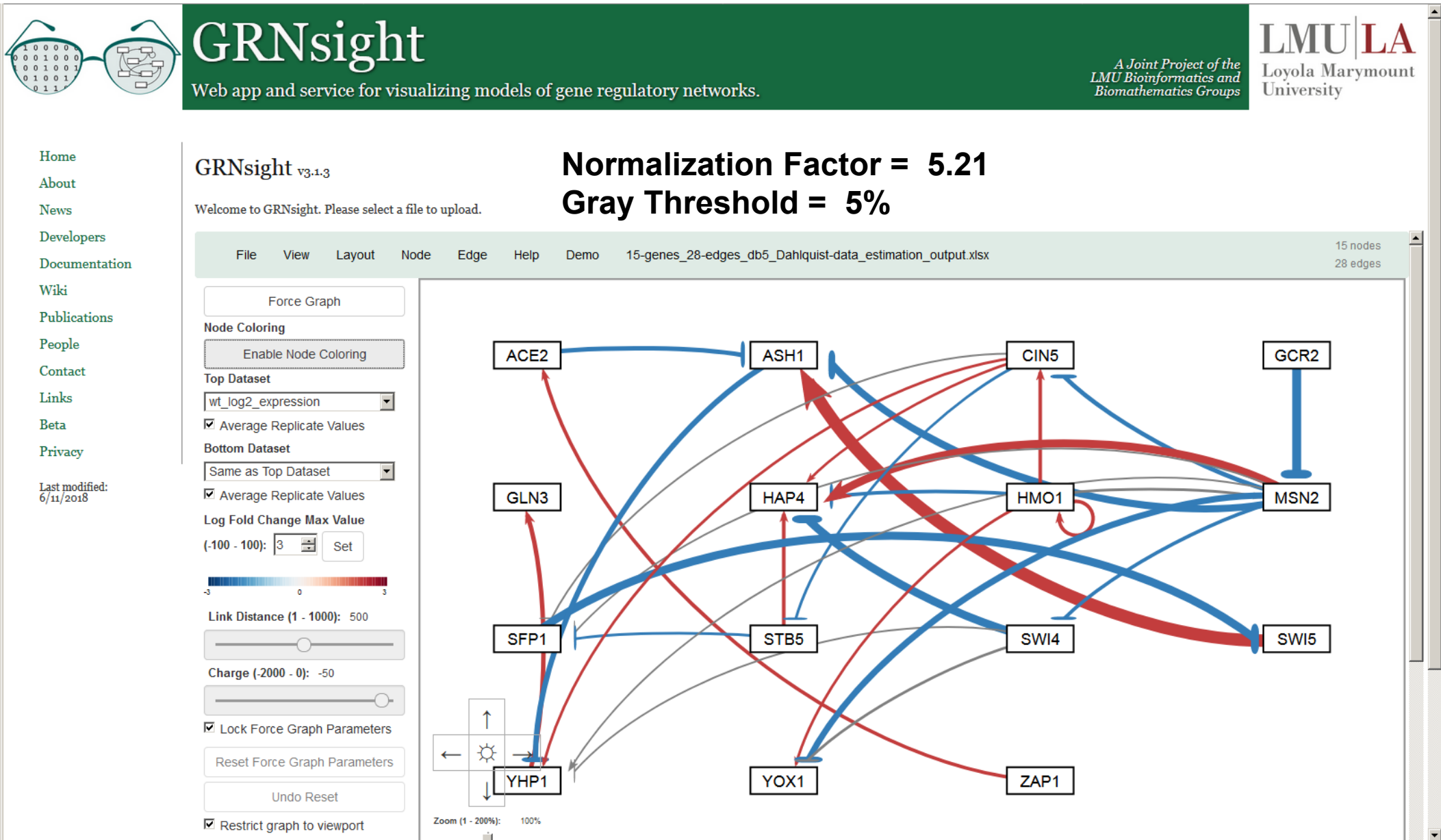
New in Version 2

And Can Be Modified to Allow Fair Visual Comparison between Different Graphs



New in Version 2

Edges with Normalized Weight Values within 5% of Zero Are Displayed as Gray to Minimize Their Visual Strength



Users Can Change the Gray Threshold with a Slider



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GRNsight v3.1.3

Welcome to GRNsight. Please select a file to upload.

Normalization Factor = 5.21

Gray Threshold = 10%

FileViewLayoutNodeEdgeHelpDemo15-genes_28-edges_db5_Dahlquist-data_estimation_output.xlsx15 nodes28 edges

Force Graph

Node Coloring

Enable Node Coloring

Top Dataset

wt_log2_expression

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value

(-100 - 100): 3Set

Link Distance (1 - 1000): 500

Charge (-2000 - 0): -50

☒ Lock Force Graph Parameters

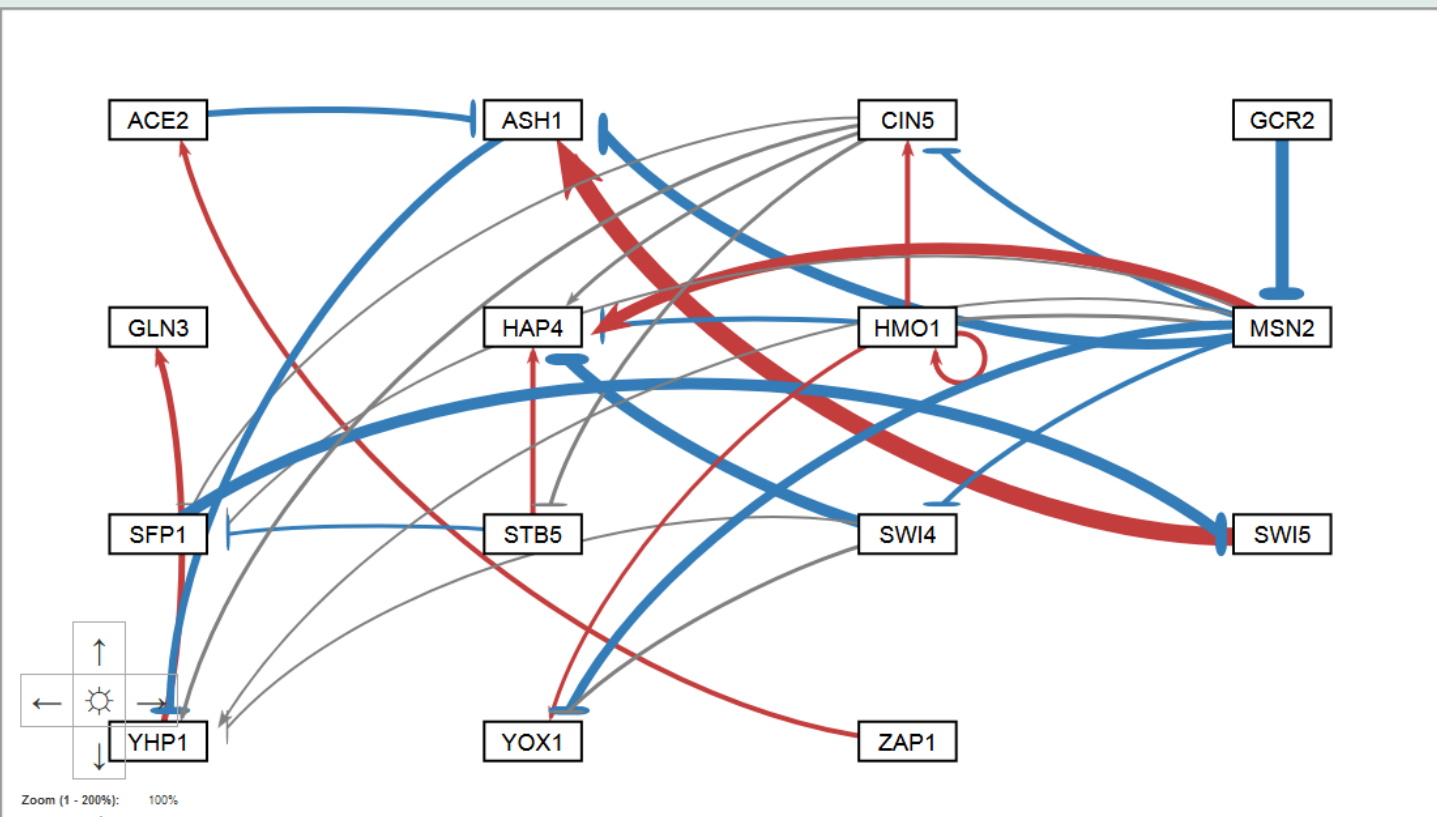
Reset Force Graph Parameters

Undo Reset

☒ Restrict graph to viewport

Viewport Size

Small (1104 X 648 pixels)



New in Version 2

Users Can Change the Gray Threshold with a Slider



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GRNsight v3.1.3

Welcome to GRNsight. Please select a file to upload.

Normalization Factor = 5.21

Gray Threshold = 25%

FileViewLayoutNodeEdgeHelpDemo15-genes_28-edges_db5_Dahlquist-data_estimation_output.xlsx15 nodes28 edges

Force Graph

Node Coloring

Enable Node Coloring

Top Dataset

wt_log2_expression

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value

(-100 - 100): 3Set

Link Distance (1 - 1000): 500

Charge (-2000 - 0): -50

☒ Lock Force Graph Parameters

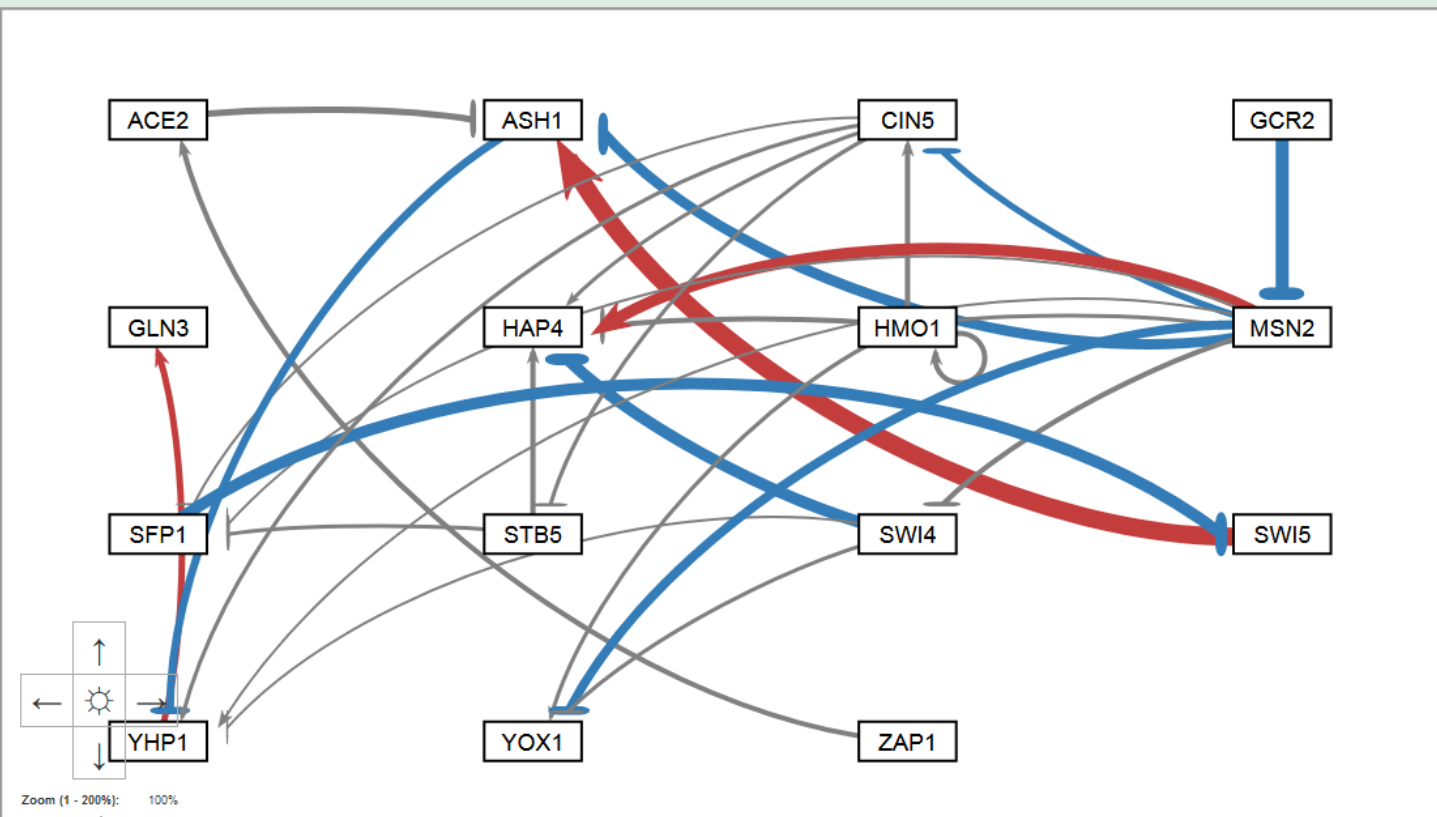
Reset Force Graph Parameters

Undo Reset

☒ Restrict graph to viewport

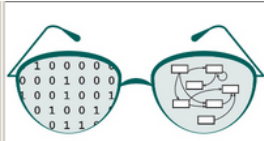
Viewport Size

Small (1104 X 648 pixels)



New in Version 2

Users Can Change the Gray Threshold with a Slider



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GRNsight v3.1.3

Welcome to GRNsight. Please select a file to upload.

Normalization Factor = 5.21

Gray Threshold = 50%

File View Layout Node Edge Help Demo 15-genes 28-edges db5 Dahlquist-data estimation output.xlsx

15 nodes
28 edges

Force Graph

Node Coloring

Enable Node Coloring

Top Dataset

wt_log2_expression

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value

$(-100 - 100):$

Link Distance (1 - 1000): 500

Charge (-2000 - 0): -50

☒ Lock Force Graph Parameters

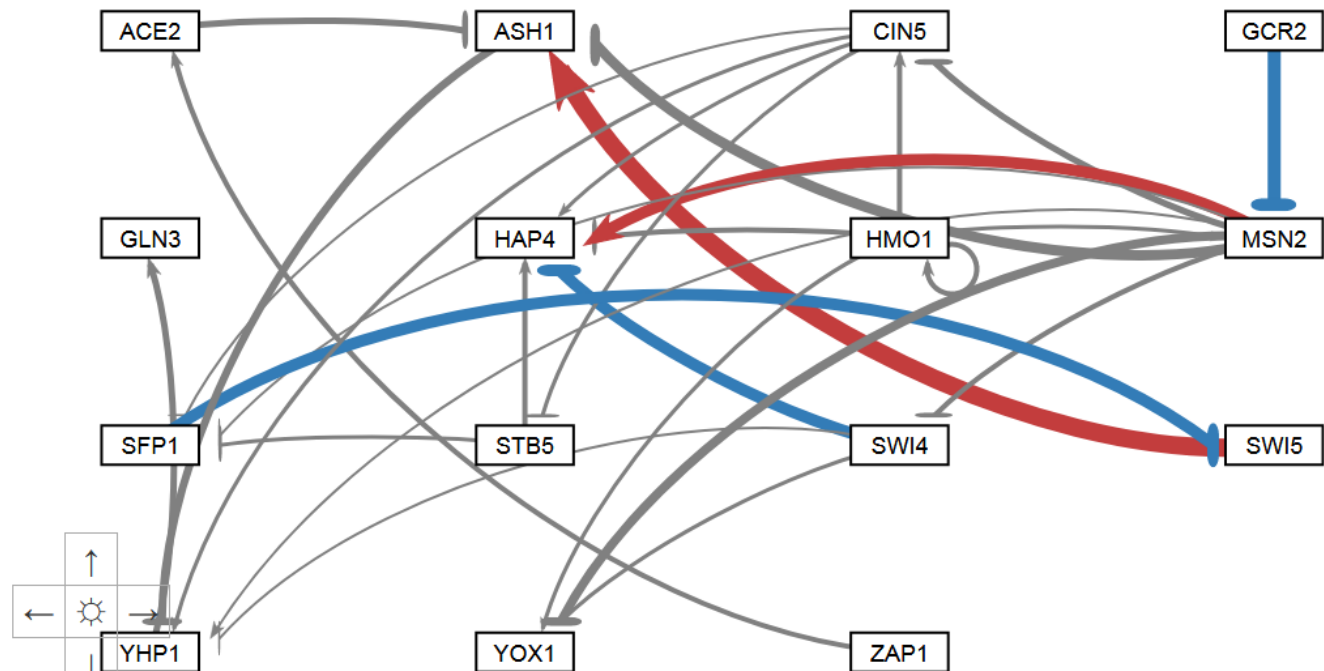
Reset Force Graph Parameters

Undo Reset

☒ Restrict graph to viewport

Viewport Size

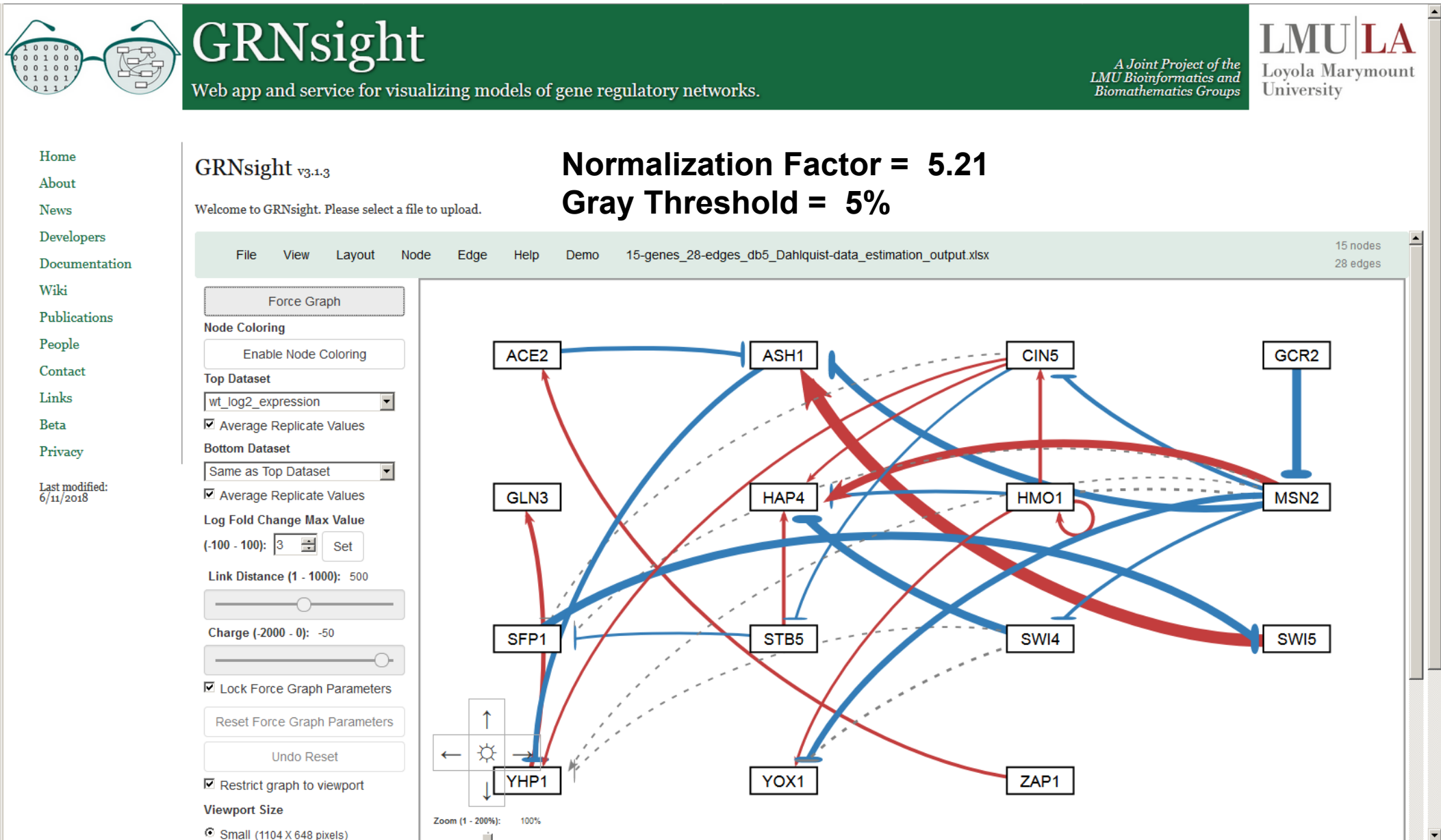
Small (1104 X 648 pixels)



Zoom (1 - 200%): 100%

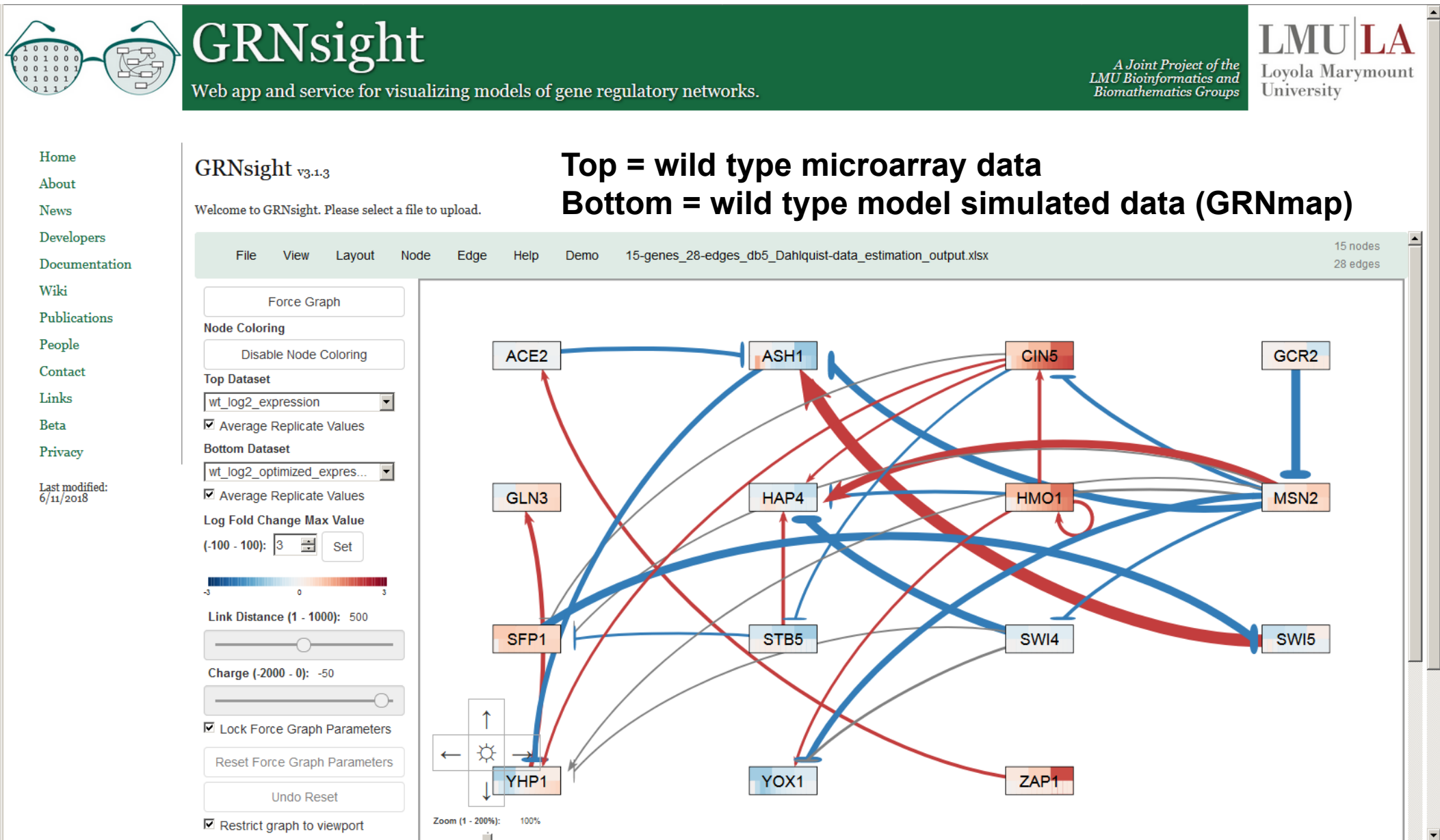
New in Version 2

Gray Edges Can Also Be Shown as Dashed Lines to Facilitate Visualization by Color Blind Users



New in Version 3

Nodes Can Display the Measured and/or Simulated Expression Data as Individual Heat Maps



Right-clicking on a Gene Opens a New Web Page with Information Dynamically Retrieved from Source Databases



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5/18/2018

GLN3

Saccharomyces cerevisiae (strain ATCC 204508 / S288c)

General Information

Protein Information

Regulation

Gene Ontology

Sources

General Information

SGD ID:	S000000842 ^[1]
NCBI Gene ID:	856763 ^[4]
Ensembl ID:	Not found ^[3]
Uniprot ID:	GLN3_YEAST ^[2]
JASPAR ID:	MA0307.1 ^[5]
Description:	Transcriptional activator of genes regulated by nitrogen catabolite repression; localization and activity regulated by quality of nitrogen source and Ure2p ^[1]
Species:	<i>Saccharomyces cerevisiae</i> (strain ATCC 204508 / S288c) ^[2]
Locus Tag:	YER040W ^[4]
JASPAR Family:	GATA-type zinc fingers ^[5]
JASPAR Class:	Other C4 zinc finger-type factors ^[5]
Chromosome Sequence:	NC_001137.3 ^[4]

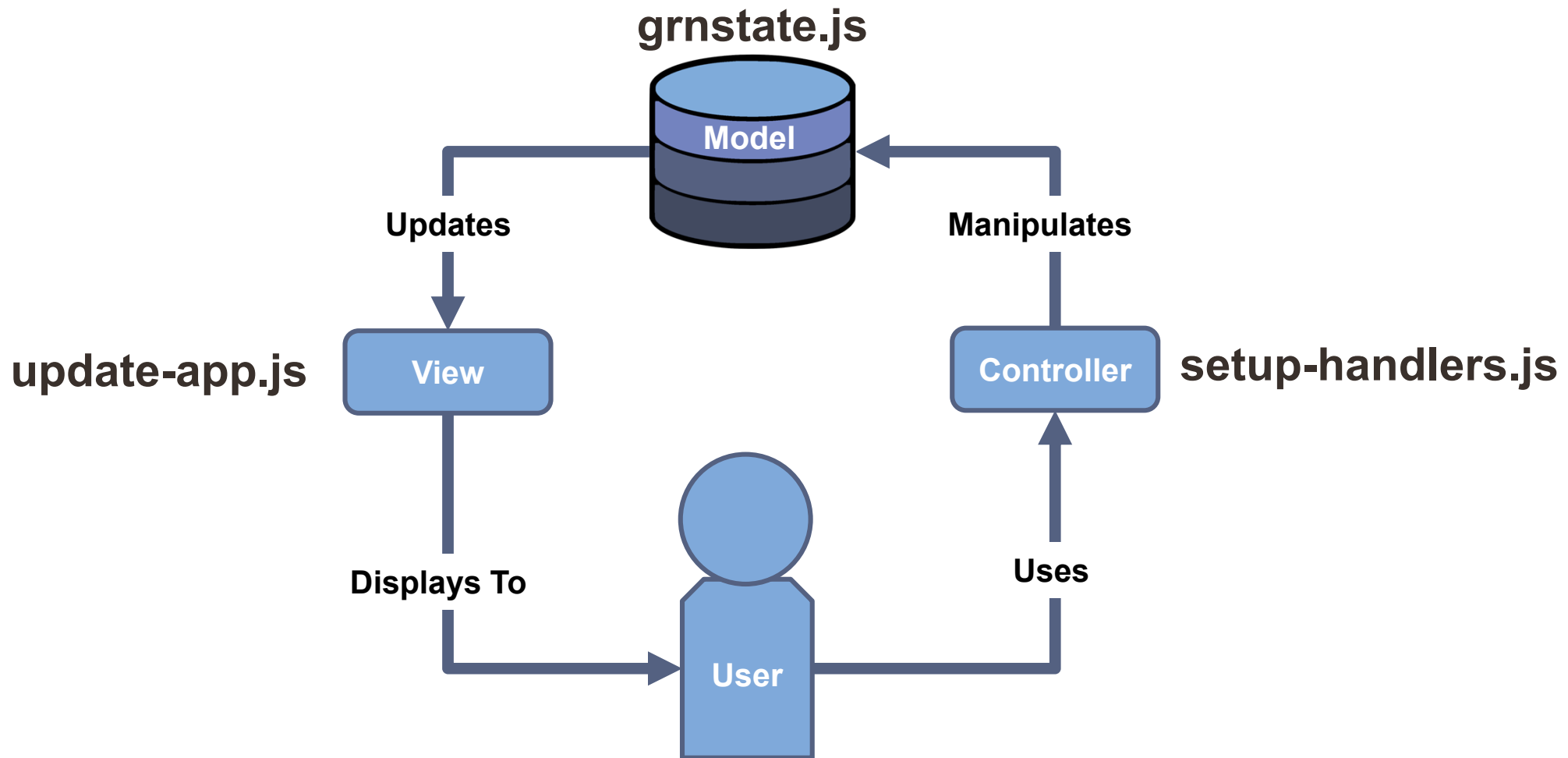
Protein Information

Regulation

Gene Ontology

Sources

Version 4 of GRNsight has been refactored to a Model-View-Controller Paradigm



12 files → 6 files

Data from a Backend PostgreSQL Expression Database Can Now Be Used to Color Nodes



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04/30/2020

Beta v4.0.18

Disclaimer: This is the beta version of GRNsight for live testing of new functionality. For the most stable version, please go to the [GRNsight home page](#).

File Layout Node Edge View Species Help Demo Demo #2: Weighted GRN (15 genes, 28 edges, Dahlquist Lab unpublished data)

15 nodes
28 edges

Layout

Node

☒ Enable Node Coloring

Select from user-uploaded
expression data, or use data
from our Expression Database

Top Dataset

Kitagawa_2002_wt

☒ Average Replicate Values

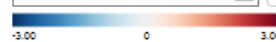
Bottom Dataset

Thorsen_2007_wt

☒ Average Replicate Values

Log Fold Change Max Value
(0.01-100):

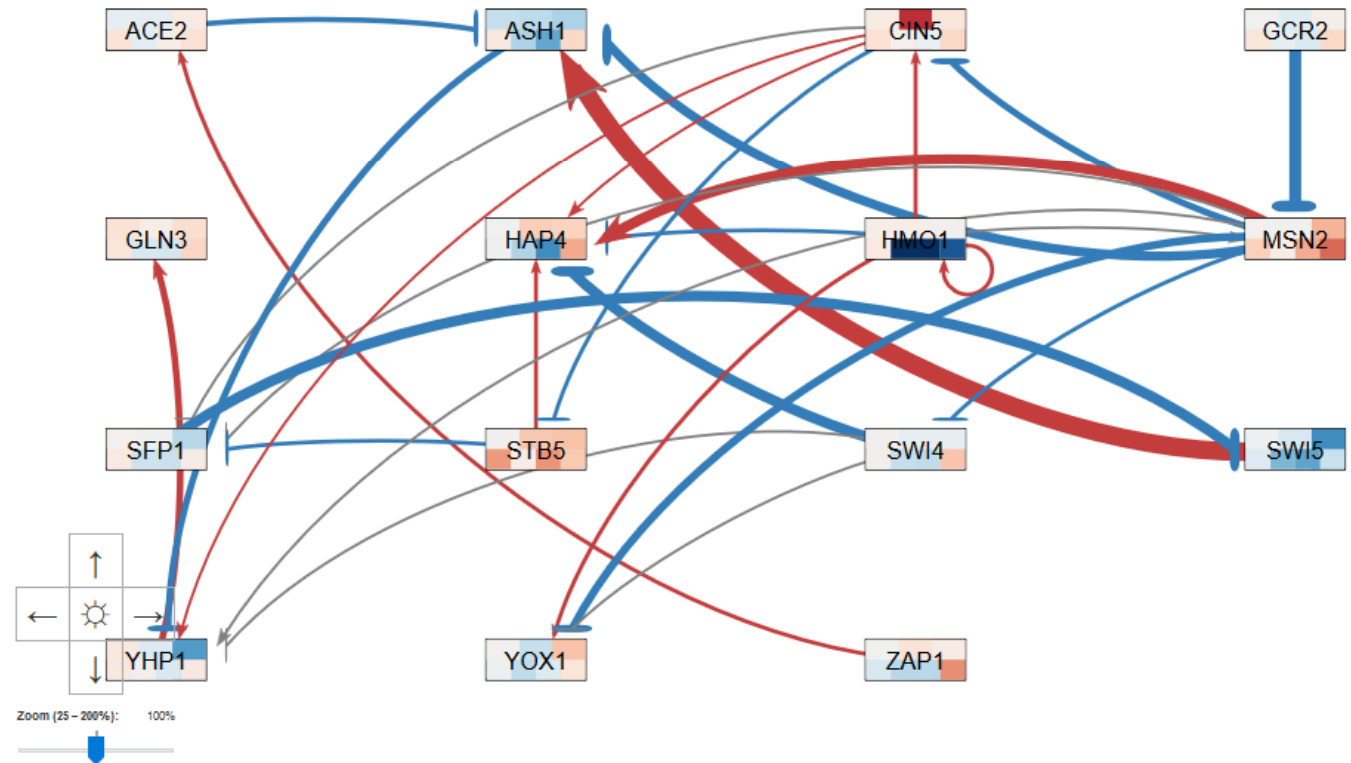
3



Edge

View

Species



New in Version 4 beta

Acknowledgments

John David N. Dionisio

Mihir Samdarshi

Nicole A. Anguiano

Anindita Varshneya

Eileen J. Choe

Yeon-Soo Shin

Alexia M. Filler

John L. Lopez

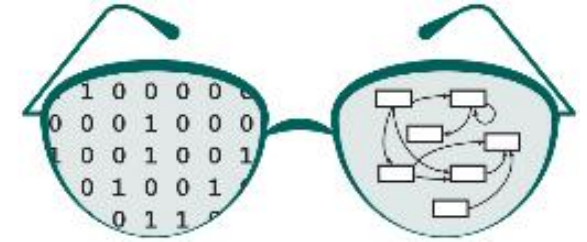
Edward B. Bachoura

Justin Kyle T. Torres

Kevin B. Patterson

Ona O. Igbinedion

BIOL 367: Biological Databases, Fall 2017 and Fall 2019



LMU Kadner-Pitts Research Grant

LMU Honors Program

LMU Rains Research Assistant Program

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<http://dondi.github.io/GRNsight/>



<https://github.com/dondi/GRNsight>